

Chamber Geometry Design for Small Bipropellant Liquid Rocket Engine

Truman Abbe

July 2026

1 Design

Fuel: ethanol.

Oxidizer: nitrous oxide.

Fuel to oxidizer mass ratio: 3 to 1 (Oxidizer to Fuel). This is fuel rich to reduce peak combustion temperatures.

Target max thrust: A) 30 N (6.74 lb). Estes D12-7 has max thrust of 29.7 N. B) 50 N (11.24 lb). This increases the nozzle throat diameter, lowering the contraction ratio. This thrust target is used for design candidate 6.

Chamber material: mild steel. This is cheaper than stainless steel.

The model is constructed using the following governing equations. Each equation is numbered and accompanied by a brief explanation that defines each variable and its corresponding MATLAB variable.

1. Throat Diameter

Variable definition: D_t is the diameter at the nozzle throat, where the flow reaches Mach 1.

$$D_t = \sqrt{\frac{4F}{\pi P_c C_F}} \quad (1)$$

MATLAB: $F = \mathbf{F}$ (target thrust, N), $P_c = \mathbf{P_c}$ (chamber pressure, Pa), $C_F = \mathbf{C_F}$ (thrust coefficient, dimensionless), $D_t = \mathbf{D_t}$ (throat diameter, m).

2. Throat Area

Variable definition: A_t is the cross-sectional area at the throat.

$$A_t = \frac{\pi D_t^2}{4} \quad (2)$$

MATLAB: $D_t = \mathbf{D_t}$ (throat diameter, m), $A_t = \mathbf{A_t}$ (throat area, m²).

3. Contraction Ratio

Variable definition: ϵ_c is the ratio of chamber area to throat area, representing how much the flow is accelerated into the throat.

$$\epsilon_c = \frac{\pi D_c^2}{4A_t} = \left(\frac{D_c}{D_t} \right)^2 \quad (3)$$

MATLAB: $D_c = \text{D_c}$ (chamber diameter, m), $D_t = \text{D_t}$ (throat diameter, m), $\epsilon_c = \text{eps_c}$ (contraction ratio, dimensionless).

4. Nozzle Length

Variable definition: L_n is the total length of the conical nozzle, including both the convergent and divergent sections, for a given half-angle θ . The total length is the sum of the convergent length (from chamber to throat) and the divergent length (from throat to exit).

$$L_n = \frac{D_c - D_t}{2 \tan(\theta)} + \frac{D_e - D_t}{2 \tan(\theta)} = \frac{D_c + D_e - 2D_t}{2 \tan(\theta)} \quad (4)$$

MATLAB: $D_c = \text{D_c}$ (chamber diameter, m), $D_t = \text{D_t}$ (throat diameter, m), $D_e = \text{D_e}$ (exit diameter, m), $\theta = \text{Th}$ (nozzle half-angle, deg), $L_n = \text{L_n}$ (nozzle length, m).

Note: For the special case where the exit diameter equals the chamber diameter ($D_e = D_c$), this simplifies to:

$$L_n = \frac{D_c - D_t}{\tan(\theta)} \quad (4a)$$

5. Characteristic Length

Variable definition: L_{ch} is the characteristic length, a measure of combustion residence time defined as chamber volume divided by throat area.

$$L_{ch} = L_c \epsilon_c \quad (5)$$

MATLAB: $L_c = \text{L_c}$ (chamber length, m), $\epsilon_c = \text{eps_c}$ (contraction ratio, dimensionless), $L_{ch} = \text{L_ch}$ (characteristic length, m).

6. Chamber L/D Ratio

Variable definition: LD_{rat} is the length-to-diameter ratio of the cylindrical chamber, influencing combustion stability and residence time.

$$\text{LD}_{\text{rat}} = \frac{L_c}{D_c} \quad (6)$$

MATLAB: $L_c = \text{L_c}$ (chamber length, m), $D_c = \text{D_c}$ (chamber diameter, m), $\text{LD}_{\text{rat}} = \text{LDrat}$ (L/D ratio, dimensionless).

7. Expansion Ratio

Variable definition: ϵ is the ratio of nozzle exit area to throat area, determining how much the flow expands.

$$\epsilon = \left(\frac{D_e}{D_t} \right)^2 \quad (7)$$

MATLAB: $D_e = \text{D_e}$ (exit diameter, m), $D_t = \text{D_t}$ (throat diameter, m), $\epsilon = \text{eps}$ (expansion ratio, dimensionless).

8. Exit Mach Number

Variable definition: M_e is the Mach number at the nozzle exit, solved iteratively from the isentropic area-Mach relation.

$$\frac{A_e}{A_t} = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (8)$$

MATLAB: $A_e/A_t = \text{eps}$ (expansion ratio), $\gamma = \text{ga}$ (specific heat ratio, dimensionless), $M_e = \text{M_e}$ (exit Mach number, dimensionless).

9. Exit Pressure

Variable definition: P_e is the pressure at the nozzle exit, calculated from the isentropic pressure relation.

$$P_e = P_c \left(1 + \frac{\gamma - 1}{2} M_e^2 \right)^{-\frac{\gamma}{\gamma - 1}} \quad (9)$$

MATLAB: $P_c = \text{P_c}$ (chamber pressure, Pa), $\gamma = \text{ga}$ (specific heat ratio), $M_e = \text{M_e}$ (exit Mach number), $P_e = \text{P_e}$ (exit pressure, Pa).

10. Exit-to-Ambient Pressure Ratio

Variable definition: ϵ_{ep} is the ratio of exit pressure to ambient pressure. A value > 1 indicates under-expanded flow, which prevents flow separation and ensures stable operation.

$$\epsilon_{ep} = \frac{P_e}{P_{amb}} \quad (10)$$

MATLAB: $P_e = \text{P_e}$ (exit pressure, Pa), $P_{amb} = \text{P_amb}$ (ambient pressure, Pa), $\epsilon_{ep} = \text{eps_ep}$ (exit-to-ambient pressure ratio, dimensionless).

The following configuration inputs are:

Max thrust (F): 50 N
 Chamber pressure (P_c): 1,000 kPa
 Thrust coefficient (C_F): 1.3
 Chamber diameter ?
 Chamber length ?
 Exit diameter ?
 Nozzle angle (θ): 15°
 Ambient pressure (P_{amb}): 85 kPa
 Specific heat ratio (γ): 1.27

A model will be constructed:

$$f(F, P_c, C_F, D_c, L_c, \theta, D_e, P_{amb}, \gamma) = \mathbf{H},$$

$$\mathbf{H} = [D_t, \epsilon_c, L_n, L^*, L/D, \epsilon, M_e, P_e, \epsilon_{ep}].$$

The model is constructed as follows:

$$D_t = \sqrt{\frac{4F}{\pi P_c C_F}}.$$

$$\epsilon_c = \frac{D_c^2 \pi}{4A_t},$$

where

$$A_t = \frac{\pi D_t^2}{4}.$$

And,

$$L_n = \frac{D_c - D_t}{\tan(\theta)}.$$

Also, the characteristic length and length-to-diameter ratio are:

$$L^* = L_c \epsilon_c, \quad L/D = \frac{L_c}{D_c}.$$

The expansion ratio is defined as:

$$\epsilon = \left(\frac{D_e}{D_t} \right)^2.$$

The exit Mach number M_e is found numerically by solving the area-Mach relation:

$$\epsilon = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}.$$

The exit pressure and the exit-to-ambient pressure ratio are evaluated using isentropic flow relations:

$$P_e = P_c \left(1 + \frac{\gamma - 1}{2} M_e^2 \right)^{-\frac{\gamma}{\gamma - 1}},$$

$$\epsilon_{ep} = \frac{P_e}{P_{amb}}.$$

The output is $\mathbf{H} = [D_t, \epsilon_c, L_n, L^*, L/D, \epsilon, M_e, P_e, \epsilon_{ep}]$.

Recommended Candidate Geometries

All candidates use the fixed inputs:

$$F = 30 \text{ N}, \quad P_c = 1000 \text{ kPa}, \quad C_F = 1.3, \quad \theta = 15^\circ$$

which yield $D_t = 5.42 \text{ mm}$ (drill 5.5 mm) for every case.

Candidate 1 (20mm Diameter)

Inputs	Values
Chamber diameter D_c	20.0 mm (0.020 m)
Chamber length L_c	80.0 mm (0.080 m)
Outputs	Values
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	13.6
Nozzle length L_n	54.4 mm
Characteristic length L^*	1.09 m
Exit diameter $D_e = D_c$	20.0 mm

Candidate 2 (18mm Diameter)

Inputs	Values
Chamber diameter D_c	18.0 mm (0.018 m)
Chamber length L_c	90.0 mm (0.090 m)
Outputs	Values
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	11.0
Nozzle length L_n	46.9 mm
Characteristic length L^*	0.99 m
Exit diameter $D_e = D_c$	18.0 mm

Candidate 3 (22mm Diameter)

Inputs	Values
Chamber diameter D_c	22.0 mm (0.022 m)
Chamber length L_c	60.0 mm (0.060 m)
Outputs	Values
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	16.5
Nozzle length L_n	61.8 mm
Characteristic length L^*	0.99 m
Exit diameter $D_e = D_c$	22.0 mm

Candidate 4 (20mm Diameter, 100mm Length)

Inputs	Values
Chamber diameter D_c	20.0 mm (0.020 m)
Chamber length L_c	100.0 mm (0.100 m)
Outputs	Values
Throat diameter D_t	5.42 mm
Contraction ratio ϵ_c	13.6
Nozzle length L_n	54.4 mm
Characteristic length L^*	1.36 m
Exit diameter $D_e = D_c$	20.0 mm
L/D ratio	5.0

Candidate 5 (20mm Diameter, 60mm Length, 50N Thrust)

Inputs	Values
Chamber diameter D_c	20.0 mm (0.020 m)
Chamber length L_c	60.0 mm (0.060 m)
Max thrust F	50.0 N
Chamber pressure P_c	1000.0 kPa
Thrust coefficient C_F	1.300
Nozzle angle θ	15.0°
Outputs	Values
Throat diameter D_t	6.998 mm
Contraction ratio ϵ_c	8.168
Nozzle length L_n	48.524 mm
Characteristic length L^*	0.490 m
Exit diameter $D_e = D_c$	20.0 mm
L/D ratio	3.000

Justification: The target thrust is increased from 30 N to 50 N, which increases the throat diameter from 5.42 mm to 6.998 mm and lowers the contraction ratio from 13.6 to 8.168. This falls within the recommended range ($\epsilon_c = 8$ –10)

for small bipropellant engines operating at sea level. The chamber length is decreased from 100 mm to 60 mm, providing a L/D ratio of 3.0. The characteristic length is $L^* = 0.490$ m.

2 Candidate 6

Model Input Summary

Symbol	Physical Meaning	MATLAB Variable	Value
F	Target thrust	F	50 N
P_c	Chamber pressure	P_c	1,000 kPa
C_F	Thrust coefficient	C_F	1.3
D_c	Chamber diameter	D_c	20 mm
L_c	Chamber length	L_c	100 mm
θ	Nozzle half-angle	Th	15 deg
D_e	Exit diameter	D_e	10 mm
P_{amb}	Ambient pressure	P_amb	85 kPa
γ	Specific heat ratio	ga	1.27

Model Output Summary

Symbol	Physical Meaning	MATLAB Variable	Value
D_t	Throat diameter	D_t	6.998 mm
ϵ_c	Contraction ratio	eps_c	8.168
L_n	Nozzle length	L_n	29.86 mm
L_{ch}	Characteristic length	L_ch	0.817 m
LD _{rat}	L/D ratio	LDrat	5.0
ϵ	Expansion ratio	eps	2.042
M_e	Exit Mach number	M_e	2.124
P_e	Exit pressure	P_e	106.727 kPa
ϵ_{ep}	Exit-to-ambient pressure ratio	eps_ep	1.256

Design Criteria Check

- Characteristic length: $L_{ch} = 0.817$ m (recommended range: 0.75–1.25 m) ✓
- L/D ratio: LD_{rat} = 5.0 (recommended range: 4–6) ✓
- Contraction ratio: $\epsilon_c = 8.168$ (recommended range: 2.5–10) ✓
- Exit pressure ratio: $\epsilon_{ep} = 1.256$ (recommended range: 1.0–1.5) ✓